

Structure Learning of Agency: Recovering Interpretable Generative Models from Behaviour with Tensor Networks

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Abstract. This paper is the sequel to our study of agency in AI systems, which set out the properties an agentic system should have and used empowerment to gauge how much control an agent holds over its own future. That study measured models we built by hand; fitting a model to behaviour, the other half of computational phenotyping, was left as a plan. We carry it out. A matrix product state, an interpretable tensor network, is trained unsupervised on action–observation rollouts of a T-maze, so its generative model $p(o, a)$ is learned from behaviour rather than written down by us. We then read off the trained network several kinds of structure that a plain density estimate leaves hidden. Normalised hidden states come first, recovered by a gauge-invariant criterion of predictive equivalence, where earlier tensor-network work found the bonds “not normalised and therefore not usable” and stopped. From the states we rebuild the labelled likelihood $A = p(o \mid s)$ and transition $B = p(s' \mid s, a)$, which match the environment to total-variation 0.000 and reproduce its branching exactly; the conditional-independence graph, from exact mutual information; and the observation’s factorisation, in which context appears as a near-independent modality. The number of hidden states is fixed by a model-selection argument, read as active-inference model expansion and Bayesian model reduction rather than off a truncation threshold. Empowerment computed on the recovered model tracks the three control regimes of the first paper to three decimals, and rises when the agent acts to resolve its uncertainty even though nothing in training rewards that. Since the tensor network is not itself an active-inference agent, the method carries over to any system whose sensorimotor stream can be modelled.

Keywords: Active inference · Agency · Computational phenotyping · Empowerment · Tensor networks · Structure learning · World models

1 Introduction

Agency in artificial systems is asserted far more often than it is measured. Our first paper [1] (Paper I) held that a useful notion of agency turns on three properties, intentionality, rationality, and explainability, and argued that active inference [2] supplies all three at once: within a partially observable Markov decision process (POMDP) the agent’s posteriors, its prior preferences, and its expected-free-energy-driven policy give beliefs, desires, and a normative reason for acting, each inspectable in the likelihood and transition tensors. On such a model, empowerment [3,4], the channel capacity from an action to the next observation, measures how much of what the agent will observe its actions can actually decide. This paper is the follow-up. It keeps the definition, the probe, and the T-maze of Paper I fixed, and adds the step Paper I set aside.

Phenotyping, taken from computational psychiatry [5,6], runs in two steps: fit a generative model to behaviour, then read behaviour off the fitted parameters. Paper I did only the second, on models we specified by hand, and described the first as work to come. We do it here. Once the model is learned from behaviour, the reading that produces an empowerment number also produces the entire generative model that stands behind it.

Concretely, we train a *matrix product state* (MPS), a one-dimensional tensor network, as an unsupervised generative model of T-maze rollouts, following Wauthier et al. [7]. Tensor networks are attractive here because they admit exact marginalisation: the conditionals the empowerment probe needs, and the mutual informations a dependency analysis needs, are available in closed form from reduced density matrices. But a trained density model is not yet a *structured* one. Wauthier et al. observed that although the bonds between tensors “can be regarded as internal states, they are not normalised and therefore not usable,” and consequently bypassed hidden-state recovery entirely. Our central technical contribution is to supply exactly that missing piece and to build a full structural readout on top of it.

Contributions. From MPS models trained on the minimal four-observation T-maze and the full pymdp T-maze (24 composite observations), we recover, and validate against analytic ground truth:

1. **Hidden states** by gauge-invariant fidelity clustering, corrected to a *predictive-equivalence* criterion that is robust to gauge coherences the model writes into the bond (Sec. 3);
2. **A labelled generative model**: emission $A = p(o \mid s)$ and transition $B = p(s' \mid s, a)$, matched to the environment up to a state permutation (Sec. 4);
3. the **conditional-independence graph** over observables, by exact mutual information (Sec. 5);
4. the **observation factorisation**, from subsystem mutual information of the single-site reduced density matrix (Sec. 6);
5. the **number of hidden states** by model selection, framed as active inference model expansion and Bayesian model reduction (Sec. 7); and

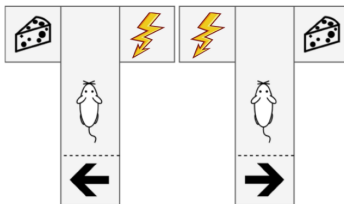


Fig. 1. The T-maze. The arrow marks the observation revealing where the cheese is; entering an arm is an absorbing trap, so the epistemically optimal first move is the cue.

6. **symbolic states:** a gauge that makes the bond index itself a labelled hidden state (Sec. 8).

On the learned model we then phenotype control capacity (Sec. 9). All code accompanies the paper.

2 Background

We recap only what the pipeline needs; the three agency criteria, the empowerment probe, and the design of the T-maze are developed in full in Paper I [1], and the tensor-network machinery is new to this paper.

2.1 The task and the analytic model

The minimal T-maze is a two-step POMDP with three actions (Left, Right, Cue) and four observations (Cheese, Shock, Left, Right); cheese and shock are placed left/right with probability $1/2$ and the cue reveals their location, while entering an arm is an absorbing trap (Fig. 1) [1]. The full `pymdp` [8] T-maze factors the observation into Position (4), Reward (3), and Context (2), yielding 24 composite observations over a three-step episode beginning with a null action. Both admit an exact generative model $q(o, a)$, which we use only as ground truth for validation—the learner never sees it.

2.2 MPS as a generative model

An MPS represents the amplitude of a length- L sequence as a chain of contracted tensors, one per time step, each carrying a physical index (here the pair [action, observation]) and two bond indices. Trained by the Born rule $p(x) = |\psi(x)|^2/Z$ [9], it is a *Born machine*, and admits exact contraction of any marginal or conditional. Non-negative MPS are exactly hidden Markov models, with the bond index a classical hidden state [10]; a generic complex Born machine is strictly more expressive and has no such exact form, so hidden states must be recovered rather than read off (Sec. 3). We train each model with a DMRG-style cumulant gradient sweep and a learning-rate schedule that shrinks the step when the loss stalls [9]; the minimal model reaches $\|p - q\|_1 = 0.13$ and the full model $\|p - q\|_1 = 0.005$.

2.3 Empowerment

We borrow one quantity from Paper I, $\max_{p(a)} I(A; O')$, the largest mutual information an agent can drive between its next action and its next observation, evaluated by Blahut–Arimoto [4]. Paper I distinguishes the value computed under the true environment (*objective*) from the value under the agent’s beliefs (*subjective*), and both from the *actual* value once preferences narrow the action distribution. Only one variant is empirical here: computed under the learned model, the subjective value approaches the objective one as the fit tightens. We report it over the full observation and over the reward modality alone.

3 Recovering hidden states

The bond of a trained MPS is defined only up to a gauge transformation $A^{(n)} \rightarrow A^{(n)}G$, $A^{(n+1)} \rightarrow G^{-1}A^{(n+1)}$, so the bare bond index has no intrinsic labelling, so any state we read off it must be invariant to that freedom. Each history h (a prefix of actions and observations) drives the network into a bond vector $|b_h\rangle$; the quantity $|\langle b_i | b_j \rangle|^2$ is gauge-invariant, and grouping histories by it gives a first partition into states (Fig. 2, left/centre).

On the minimal maze this yields exactly four states (cheese-trap, shock-trap, and one per cue context), merging the two arms of entry because position is unobservable there. On the full maze, however, raw fidelity clustering *overseparates*: the model writes the uninformative context bit of the mid-step observation into the bond as a coherence that does not affect the future, splitting otherwise-identical states. The robust criterion is *predictive equivalence*—the predictive-state / observable-operator definition of state [11,12]: two histories are the same state iff they induce the same future conditional $p(o' | a', h)$. Clustering the action-conditioned predictive signatures, and cutting the dendrogram at its largest gap (Fig. 2, right), collapses the gauge duplicates and returns the correct seven states (unresolved start, four arm–outcome traps, two cue contexts). What the learned latent encodes is therefore distinguishability rather than raw history: whether the agent came from the left or the right arm collapses into one state when position cannot be seen, and splits into two once it can.

4 A labelled generative model: emission and transition

Given the states, we reconstruct the factorised model. Because probabilities are Born-squared, we push amplitudes to probabilities *before* forming any stochastic object: attempting to read a stochastic matrix off the complex tensor, or to gauge-transform the model into a non-negative form, is either ill-defined or NP-hard for a genuine Born machine [10]. The **emission** $A = p(o | s)$ is estimated by conditioning the exact joint on each state’s histories and marginalising the action; the **transition** $B = p(s' | s, a)$ by re-contracting the bond after appending (a, o) , classifying the resulting state by predictive equivalence, and accumulating Born

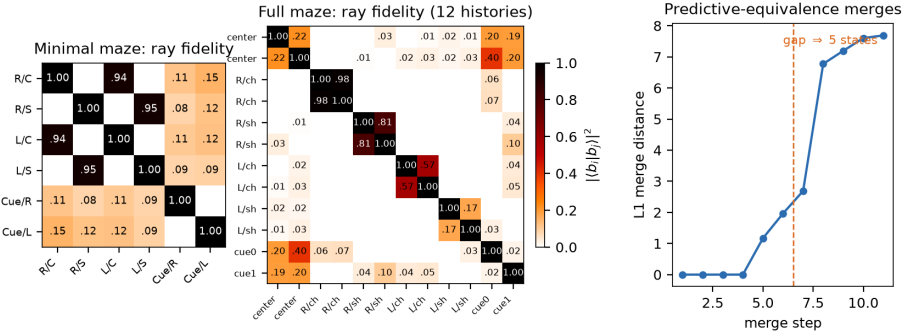


Fig. 2. State recovery. Pairwise bond-ray fidelity for the minimal maze (left, four blocks) and full maze (centre); raw fidelity over-separates on the full maze because the uninformative context bit is written into the bond. Right: predictive-equivalence merge distances—the near-zero merges collapse the gauge duplicates and the gap fixes the count at seven.

weight. States are matched to the environment by the Hungarian algorithm, the only residual gauge freedom being a discrete permutation.

The reconstruction matches ground truth to measurement precision (Fig. 3, Table 1). On the minimal maze the emission matches ground truth to total-variation 0.000 for all four states. On the full maze, all seven emissions match to mean total-variation 0.000: the four traps are deterministic, and the two cue states carry the correct *opposite* action-conditioned reward patterns (under Right, context 0 yields cheese and context 1 yields shock). The recovered transition reproduces the branching structure of the environment exactly: from the start, Right and Left each split 0.50/0.50 over their two arm-outcome states, Cue splits 0.50/0.50 over the two context states, and staying at the centre is deterministic. This is the object Wauthier et al. could not extract: a normalised, labelled, ground-truth-comparable A and B read from a network trained on behaviour alone.

5 The dependency graph

Because the learned joint is available in closed form, its dependency structure follows from exact mutual information. We form the pairwise MI matrix over the observed variables and take its maximum-weight spanning tree (Chow-Liu [13]), the tensor-network analogue of reading correlation structure off a quantum state [14] (Fig. 4). The recovered tree is the correct temporal chain: the action at each step is strongly coupled to its observation and the observation to the next. Conditional mutual information supplies exact independence certificates: $I(a_1; o_3 \mid o_2) = 0$ (the first action affects the last observation only through the intervening one) and, in the minimal maze, $I(a_1; o_3 \mid o_2) = 0$ with $I(o_2; o_3 \mid a_2) = 1.31$ bits (dependence where the model should be dependent).

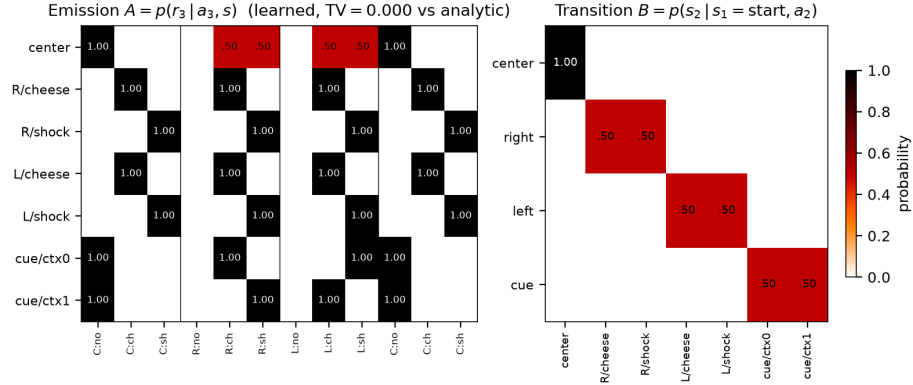


Fig. 3. Recovered generative model (full maze). Left: emission $A = p(r_3 | a_3, s)$ over seven states and four actions $\times$ three rewards; each state’s pattern matches analytic ground truth (mean TV 0.000). Right: transition $B = p(s_2 | s_1 = \text{start}, a_2)$, recovering the exact 0.50/0.50 arm and cue branching.

We validate against the *observable-projected* ground-truth graph, since marginalising the latent state induces real dependencies among observables that a naive comparison to the full dynamic Bayes net would misread as error.

6 Discovering the observation factorisation

The full-maze model is fed a *flat* 24-dimensional observation; the Position $\times$ Reward $\times$ Context factorisation is assumed only in analysis. We test whether the learned tensors respect it, from the single-site reduced density matrix $\rho = \text{Tr}_{\text{rest}} |\psi\rangle\langle\psi|$. Subsystem quantum mutual information $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ vanishes iff the factors are independent [15]. The model discovers context as a *near-independent* modality ($I(\text{pos:ctx}), I(\text{rew:ctx}) \approx 0.05$ bits) while position and reward remain coupled (1.0 bit)—which is causally correct, since position gates whether a reward is possible (Fig. 4, right). A search over groupings of the 24 indices confirms it: the trace-distance residual $\frac{1}{2}\|\rho - \bigotimes_k \rho_k\|_1$ is far lower for (position, reward) $\times$ context (0.21) than for full separation (0.54). The model organises the observation along the environment’s causal axes without being told they exist.

7 Selecting the number of states

That the recovered dimensionality is a *model-selection* result, not an artefact of an SVD threshold, is shown by sweeping the capped bond dimension χ and training each model to convergence (Fig. 5). On the minimal maze the fit improves sharply until $\chi = 4$ (fit L1 $0.17 \rightarrow 0.03$) and then goes flat: extra bond dimension is unused “spare capacity.” We read this through active inference structure

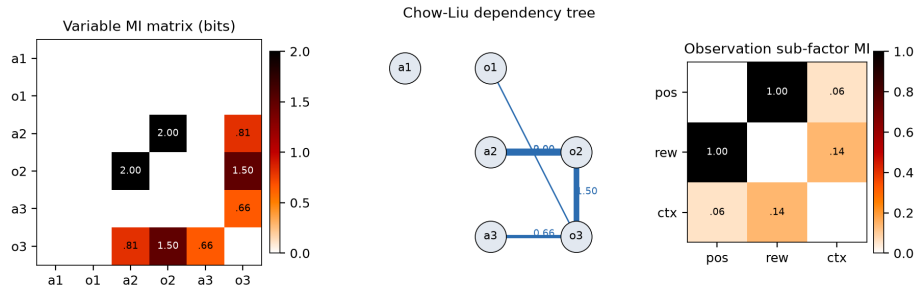


Fig. 4. Dependency and factorisation (full maze). Left: pairwise mutual-information matrix over the six observed variables. Centre: the Chow-Liu tree recovers the temporal action-observation chain. Right: subsystem mutual information of the single-site reduced density matrix—the model discovers context as a near-independent factor ($I(\text{pos:ctx}), I(\text{rew:ctx}) \approx 0.05$) while position and reward are coupled (1.0 bit), the causally correct structure.

learning [16,17]: a large- χ model has spare capacity; training commits capacity to states that raise model evidence (accuracy gain exceeding complexity cost) and leaves the rest unused, so the selected count is the fixed point of a free-energy trade-off over model structure—model expansion followed by Bayesian model reduction. A description-length penalty is one state more conservative here, because the fourth state adds interpretable low-probability structure that the fit needs but the likelihood barely rewards; the L_1 knee is the decisive signal.

8 Symbolic states by gauge fixing

Finally we remove the post-hoc character of state extraction by choosing the residual gauge that rotates the bond into the cluster basis. From the k cluster representatives we build an orthonormal frame (the eigenbasis of the history-aggregated bond density is more robust than a raw QR) and complete it to a unitary U ; applying U as the gauge preserves the represented distribution exactly (here to 4×10^{-14}) while the bond index becomes a labelled state that the model *emits*, $p_i(h) = |\langle q_i | U^\dagger | b_h \rangle|^2$ (Fig. 6, right) [18,19]. On the minimal maze the emitted distributions agree with the fidelity clustering (adjusted Rand index 1.0), near one-hot for the trap states and softer for the cue states—the expected signature of a genuinely complex Born machine, for which an exact classical-state gauge does not exist [10], quantified rather than hidden.

9 Phenotyping control capacity on the learned model

With the model recovered, we phenotype it. Subjective potential empowerment computed from the converged full-maze network reproduces the analytic regimes to three decimals (Fig. 6, left; Table 1): 2.00 bits at the start, 0.00 in the trap,

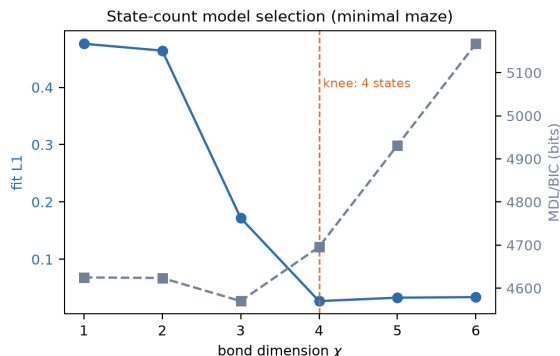


Fig. 5. State-count selection (minimal maze). Fit L1 (blue) drops $\sim 6\times$ from $\chi = 3$ to $\chi = 4$ and then plateaus; bond dimension beyond four is spare capacity. The description-length score (grey) is one state more conservative.

Table 1. Structure recovery against analytic ground truth. Emission/transition are total-variation distances; empowerment is bits (full/reward). “Learned” is the converged full-maze MPS ($\|p - q\|_1 = 0.005$).

Quantity	Analytic	Learned	Error
Emission $A = p(o s)$ (mean TV over 7 states)	—	—	0.000
Transition B branching (Right $\rightarrow$ {cheese,shock})	0.50/0.50	0.51/0.49	≤ 0.01
Empowerment, start	2.00/1.00	2.00/1.00	0.00
Empowerment, trap	0.00/0.00	0.00/0.00	0.00
Empowerment, post-cue	2.00/1.585	2.00/1.585	< 0.001
Hidden-state count (minimal / full)	4 / 7	4 / 7	exact

and 2.00 bits (full) / 1.585 bits (reward) post-cue, against analytic 2.00/1.585. Restricting the probe to the reward modality returns the three control regimes of the minimal maze (1.00/0.00/1.585 bits): reading empowerment only over outcomes the agent cares about isolates the control that matters for behaviour, which the full observation space dilutes. The subjective–objective gap is now empirical and small: a bond-truncated checkpoint attenuates post-cue empowerment (0.65 bits reward-only), and it approaches the objective value as the fit improves. Acting to resolve uncertainty increases empowerment: visiting the cue lifts it from the intermediate to the maximal regime, though empowerment is no part of the training objective.

10 Discussion

What is new. Earlier tensor-network work learned $p(o, a)$ but could not expose normalised hidden states, and the phenotyping programme it was meant to serve was demonstrated only on hand-specified models. We supply the missing state

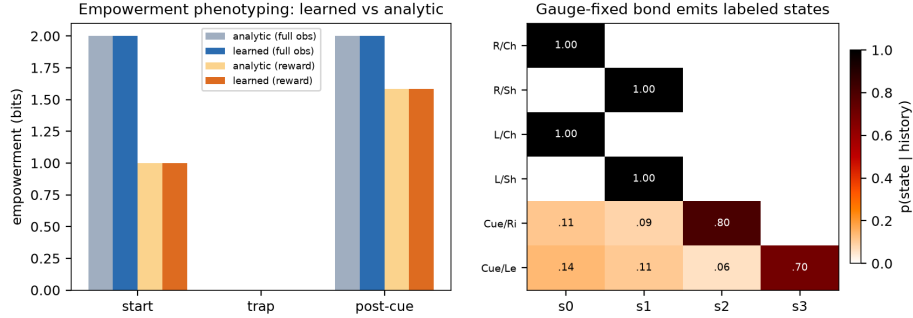


Fig. 6. Left: empowerment phenotyping on the learned full-maze model matches analytic across the zero-, intermediate-, and maximal-control regimes, full-observation and reward-only. Right: the gauge-fixed bond emits labelled states (minimal maze); trap histories are near one-hot, cue histories softer.

construction, gauge-invariant and based on predictive equivalence, and build on it a complete structural account: a labelled likelihood and transition validated to $\sim 10^{-3}$, the conditional-independence graph, the observation factorisation, a principled state count, and a symbolic-state gauge. Agency phenotyping then runs on the learned model and matches the analytic regimes. The pipeline is architecture-agnostic: the tensor network is not an active-inference agent, so recovering active-inference-style structure from it addresses the circularity worry that active inference might satisfy the agency criteria only by construction.

Senses of structure learning. We are precise about the phrase. The model performs unsupervised discovery of latent dimensionality (Sec. 7), of the task-relevant state abstraction (Sec. 3), of the dependency graph (Sec. 5), and of the observation factorisation (Sec. 6), and it yields a labelled factored model (Sec. 4). It does not search over network topology; the tensor-network structure-search literature [20,21] would be the route to that, and is a natural next step on richer environments.

Limitations. Results are on two small POMDPs with known ground truth, so validation is against an analytic model rather than blind; a state-count criterion stated before inspection, and a held-out protocol on data an exact model cannot memorise, would strengthen the claim. Empowerment measures controllability, not agency: reading control regimes as agency profiles is licensed only because the agentic action chain is independently inspectable in the recovered model [1]. Finally, the description-length criterion for state count is conservative under our naive parameter count, which ignores gauge freedom; a gauge-aware count is future work.

Governance. Because the readout is architecture-agnostic and needs only a learnable model of the sensorimotor stream, it offers a concrete route to au-

ditioning opaque systems: fit an interpretable surrogate, recover its states and its controllability profile, and test whether governance handles track the recovered regimes [1]. Whether real interventions map onto these regimes is the empirical programme this pipeline makes testable.

11 Conclusion

Computational phenotyping of agency can be carried out end-to-end: from behaviour to a learned generative model, from that model to its recovered structure, and from the structure to a controllability phenotype that reproduces ground truth. The tensor network supplies exact conditionals and, once gauge-invariant states are fixed, a fully interpretable factored model—the inspectable surrogate that phenotyping of opaque systems requires.

Data and code. All code, models, and figures accompany the paper at <https://github.com/philipollaswilson/Tensor-Networks-for-the-Tmaze>.

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Disclosure of Interests. The authors declare no competing interests.

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